Learning Check

1. (True or False.) A conservation law can be used without knowing all of the details about what is occurring inside a system.
2. The linear momentum of a truck will always be greater than that of a bus if
   (a) the truck's mass is larger than the bus's but its speed is the same.
   (b) the truck's speed is larger than the bus's but its mass is the same.
   (c) both its mass and its speed are larger than the bus's.
   (d) Any of the above.
3. In a collision, the total ______________ is the same before and after.
4. (True or False.) During a collision between two football players, the total linear momentum of the system cannot be zero.

ANSWERS: 1. True 2. (d) 3. linear momentum 4. False

Learning Check

1. (True or False.) The parking brake in most automobiles makes use of a lever so that the force the driver must exert when setting it is reduced. Consequently, the amount of work the driver must do is also reduced.
2. Work is done on an object when
   (a) it moves in a circle with constant speed.
   (b) it is accelerated in a straight line.
   (c) it is carried horizontally.
   (d) All of the above.
3. If a force acting on an object and the object's velocity are in opposite directions, the work done by the force is ______________.
4. (True or False.) The force of gravity cannot do work.

ANSWERS: 1. False 2. (b) 3. negative 4. False

Learning Check

1. To be able to do work, a system must have ______________.
2. Identical cars A and B are traveling down a highway, with A going two times as fast as B. The kinetic energy of A is
   (a) 2 times as large as the kinetic energy of B.
   (b) 4 times as large as the kinetic energy of B.
   (c) one-half as large as the kinetic energy of B.
   (d) the same as that of B, because they have the same mass.
3. (True or False.) When you lift an object, its gravitational potential energy decreases.
4. A hiker loads camping gear into her SUV. In the process, the rear springs of her vehicle are compressed by a factor of two over their usual length. The potential energy stored in each of the springs under this condition relative to its normal value is
   (a) half as large.
   (b) twice as large.
   (c) one-fourth as large.
   (d) four times as large.
5. As a baseball player slides into home, the player's kinetic energy is converted into ______________ energy.

ANSWERS: 1. energy 2. (b) 3. False 4. (d) 5. internal

Learning Check

1. As a skier gains speed while gliding down a slope, ______ energy is being converted into ______ energy.
2. (True or False.) Work done inside an isolated system can increase the total energy in the system.
3. (Choose the *incorrect* statement.) The total kinetic energy plus potential energy of a body
   (a) can be negative.
   (b) always remains constant if the body is falling freely.
   (c) always remains constant if friction is acting.
   (d) can remain constant even if the body's speed is decreasing.
4. Diver A jumps off a platform and is going 5 m/s when entering the water. To be going 10 m/s when entering the water, diver B would have to jump off a platform that is ______ times as high as the platform A used.

ANSWERS: 1. potential, kinetic **2.** False **3.** (c) **4.** four

Learning Check

1. In a collision that is inelastic, the total ______ after the collision is not the same as before the collision.
2. In an elastic collision between two bodies, which of the following is *not* true:
   (a) the total linear momentum of the bodies is the same before and after the collision.
   (b) the total kinetic energy of the bodies is the same before and after the collision.
   (c) the total energy (all forms) of the bodies is the same before and after the collision.
   (d) the two bodies must have equal masses.
3. (True or False.) A spacecraft can gain or lose energy by passing near a planet.

ANSWERS: 1. kinetic energy **2.** (d) **3.** True

Learning Check

1. (True or False.) When more people ride upward in an elevator, it requires more power.
2. Identical cars A and B are being driven up the same steep hill. If the power output of A is larger than that of B, what must be happening?

ANSWERS: 1. True **2.** A is going faster, accomplishing the climb in less time.

Learning Check

1. The centripetal force acting on a body moving in a circle changes its linear momentum, but it does not change its ______ momentum.
2. (True or False.) If both the linear momentum and the kinetic energy of a satellite in orbit around Earth increase, then its angular momentum must also increase.
3. Complete this comparison: Force is to ______ as linear momentum is to angular momentum.

ANSWERS: 1. angular **2.** False **3.** torque